

Comparative Analysis of Machine Learning Algorithms for Predicting Air Quality Index

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Abstract—Many scientists and researchers have been worried over the past few decades about the issue of air quality analysis and forecasting, because air pollution in the modern world has become a terrible environmental issue. There have been several health, environmental, and climatic changes owing to polluted air. The major causes leading to poor air quality are urbanization and industrialization. The major and significant air pollutants that affect air quality include NO_x , SO_x , CO, $\text{PM}_{2.5}$, and PM_{10} . Government agencies employ an Air Quality Index (AQI) to communicate to the public how contaminated the air is now or how polluted it is expected to become. The major emphasis of this work is the analysis of the aforementioned concentration of air pollutants, and classifying the different pollutant concentration levels that adversely affect the maintenance of favourable air quality based on a machine learning approach. We used the dataset, which consists of data on several types of air pollutants taken hourly, at different stations across various cities in India. The hourly data were collected from 15 January 2015 to 1 July 2020. There were 16 attributes. We employed the Decision Tree, Support Vector Machine (SVM), and Random Forest. The most accurate classification is done by a random forest classification algorithm. It outperformed the other approaches with a maximum accuracy of 74%. These results will assist to enhance present research, and guide the future.

Keywords— Air pollution, Classification, Data mining, Machine learning, Air quality index

I. INTRODUCTION

At present, air pollution is one of the world's most serious concerns. The movement of pollutants, including gases, chemicals, and other environmentally polluted material, leads to air pollution and thereby causes adverse effects. The world's primary sources of hazardous air pollution include rapid growth, urbanization, and huge industrialization [1]. Long-term exposure to air pollution causes various health problems [2] including chronic respiratory diseases [3], lung infections [4], cardiovascular conditions, and cancers in a wide variety of ways [5]. Asthma, disorientation, headaches, eye and throat discomfort might result after short-term exposure.

Dangerous airborne particulates have a direct effect on the number of deaths and a lower life expectancy. A unsatisfactory air quality exists where over 90 percent of the population live. In the World Health Report 2016, about 17,000 individuals died in Tehran in 1990 due to contaminated air [6] and more than 21,000 died in 2013. According to the statistics of the American Lung Association of the USA, the country has spent over \$37 billion a year as a cost of air pollution-related illnesses [7]. Polluted air affects

not just human health, but the existence of the entire ecosystem [8]. There are direct effects and damaging repercussions on the ecology due to contaminated air concentration. The worst outcomes of contaminated air include increased global warming, changes in temperature, depletion of ozone, and damage to plants caused by acid rain.

Different studies to measure air pollution trends and their negative impact on the earth have therefore been conducted. Several businesses, including agriculture, production, transport, and construction, contribute to environmental emissions of pollutants. Air pollution is also caused by vehicle emissions, road ducts, trash, and fossil fuels, carbon heating, smoking tobacco, power plants, and certain natural sources such as volcanic dust [9]. Worldwide, cities face harmful air pollution [10] and struggle against it. Due to different objectives, such as education, acquisition and work, and more amenities, many individuals have relocated to the city from suburbs [6]. In addition, some regions are denser than their capacity. The daily traffic on the roadways is increasing due to increasing everyday transit to the cities. They produce a range of gases that damage the environment. In 2015, for example, the State of the Environment report in China states that 78.4% of cities failed to comply with national air quality requirements [11].

According to data from the World Health Organization, China's Xinjiang was the most polluted city in the world in 2020. Moreover, India possesses 14 of 15 of the world's most contaminated cities in 2018. Recently, nations regard air control as an important and vital element [12].

This paper proposes a few models using machine learning techniques for classifying and getting observations about which air pollutant concentration levels are harmful to the environment and human health. We implemented models using a decision tree, random forest, and Support Vector Machine (SVM) to identify the best algorithm. Here, we classified the data based on the Air Quality Index (AQI).

The remainder of the paper is structured as follows. Section II, which is "Literature Review," explains the highlights of the work done so far in the above context. Section III describes the methodological approach for predicting AQI while Section IV talks about the results obtained along with the discussion. Finally, Section V concludes the manuscript with future directions.

II. LITERATURE REVIEW

Many studies have been conducted to monitor air quality using different approaches and methods. Traditionally statistical and deterministic models [13] were applied for

forecasting air quality. Statistical models used past data and experiences for the prediction of future variables [14]. For examples, the multiple linear regression model and auto-regression moving average model [15] were used to predict air quality. However, these models could not give more accurate output results because of the inability to represent nonlinear patterns [14]. The deterministic method [16] applies meteorological principles and chemical models to evaluate pollutant emission to the environment [14]. But these kinds of models were based on the theoretical hypothesis that is relevant to sophisticated prior knowledge and unreliable limited emission data. Inaccurate pollutant emission data, complex surface conditions, and an incomplete theoretical basis lead to decreased output precision [14].

As a result of technological evolution, machine learning approaches have been widely used for classifying, predicting, and forecasting air quality. Artificial Neural Network (ANN), Deep Neural Network (DNN), Support Vector Regression (SVR) method, and Hybrid methods are some commonly used machine learning techniques in related to air pollution. ANN has been widely used for many prediction purposes in different fields. More inputs can be analyzed and uncertainty and complicated nonlinear issues can be resolved using ANN [17]. Therefore, the output is more accurate than the traditional models [18]. Different ANN models have been evolved to predict air quality. Some of them are Neuro-fuzzy Neural Network [19], Multi-Layer Perception (MLP), BP Neural Network (BPNN), and Recurrent Neural Network (RNN).

Huang *et al.* [18] created a forecasting model based on BP neural network algorithms and data mining techniques to predict airborne pollutants. This model has three main steps. Firstly, they identified the relationship of air pollution factors using data mining techniques [20]. Next, they created a training sample of gathered data for neural network training. Finally, they tested the sample data to evaluate the model. Fotouhi *et al.* [6] established a model using a neuro-fuzzy network with a fuzzy clustering algorithm to predict the hourly concentration of pollutants in Tehran.

Li *et al.* developed a new, long-term multiscale neural memory network technique for the prediction of air pollutant concentration using previous data on air pollutants, timing, and meteorological data [15]. PM_{2.5} was collected hourly at 12 air quality monitoring sites in Beijing City. These data were collected to assess the performance of the model presented. When compared with the STDN, the Time Delay Neural Network model (TDNN), the Autoregressive Moving Average model (ARMA), the Support Vector Regression model (SVR), and the conventional LSTM NN model, the LSTM model appears to outperform the other models.

Hamami and Dahlan [21] proposed a neural network model using LSTM to forecast air quality in Jakarta, Indonesia. LSTM is a type of recurrent neural network (RNN) that has been used successfully in recent air quality forecasting researches. Reddy *et al.* [22] used the LSTM framework for forecasting pollution in Beijing, China. They detected the PM_{2.5} concentrations for future timestamps based on time historical pollution data and meteorological data in Beijing. Singh *et al.* [23] established a deep neural network model using a single-layer LSTM model to predict PM_{2.5} concentration based on meteorological, traffic, and pollutant concentration data. The 'VayuAnukulani' system

[24] is built using a bidirectional LSTM network to detect different air pollutants (NO₂, PM_{2.5}, PM₁₀) and predict air quality in the future timestamp.

Chaudhary *et al.* proposed a profound learning model of LSTM predicting the number of air pollutants [25]. This proposed model forecasts for stations in Delhi and Agra the pollutant concentration for the next hour, six hours, and 12 hours, based on weather, traffic data, events, and national holidays, as well as pollutants such as NO₂, SO₂, O₃, CO, PM_{2.5}, PM₁₀. The author Li Xiang created a space-time, deep-learning, air pollution prediction architecture [13]. The model postulated connections between space and time

III. METHODOLOGY

This research presents classification models to classify the data based on AQI. Fig 1 shows the steps of the methodological approach. The methodology contains steps such as data collection, data preprocessing, feature selection, and training the models.

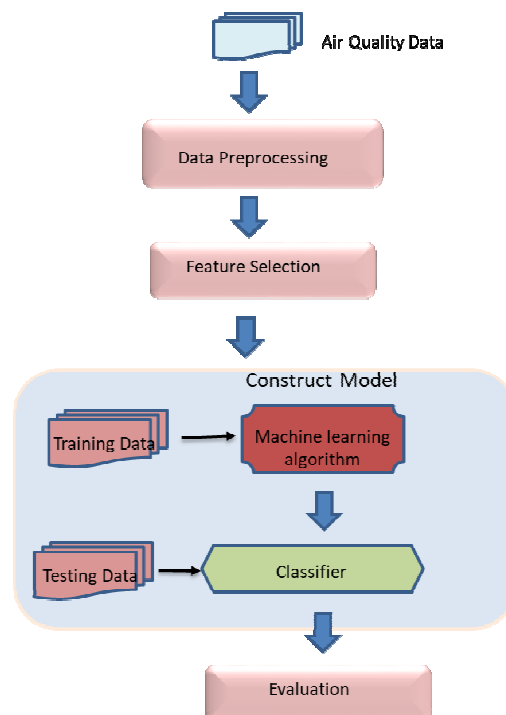


Fig. 1. Research Methodology

A. Experimental Environment

The experiments were carried out using a Windows 10 platform on a computer with an Intel Core i7-7500U, 2.70 CPU, and 8GB RAM. Weka software was used for the implementation of algorithms.

B. Data Collection

The dataset used in this research was extracted from Kaggle (<https://www.kaggle.com/>) This dataset consists of several types of air pollutants data hourly at different stations across the various cities in India. The hourly data were collected from 15 January 2015 to 1 July 2020. There were 16 attributes with 100,000 rows of data in this dataset. The dataset consists of both direct and indirect parameters related to this study. The attributes include station id, date, time, the concentrations of PM_{2.5}, PM₁₀, NO, NO₂, NO_x, NH₃, SO₂, O₃, CO, benzene, toluene, and xylene, and AQI. Particulate

Matter (PM) [26] is an air pollutant that consists of both organic and inorganic particles. Particulate varies in size. They are PM_{2.5} and PM₁₀. Both are harmful but PM_{2.5} is more dangerous than PM₁₀. PM values of other pollutant data are expressed in units of micrograms per cubic meter ($\mu\text{g}/\text{m}^3$) except for NO_x, which is measured in ppb, and CO, which is measured in mg/m^3 . Table 1 shows a sample of the dataset.

TABLE I. A SAMPLE RECORD IN DATASET

Parameter	Station Id	
	DL001	DL002
DateTime	1/1/2019 2:00	1/1/2019 1:00
PM2.5	353	361.5
PM10	538	252.5
NO	101.15	119.85
NO2	142.88	156.9
NOx	158.5	181.05
NH3	51.52	62.5
CO	2.27	3.88
SO2	22.25	18.13
O3	1.28	1.4
Benzene	3.85	6.08
Toluene	88.48	104.15
Xylene	3.13	7.75
AQI	427	427
AQI_Bucket	Severe	Severe

C. Data Preprocessing

The dataset contains some unnecessary values and attributes. Therefore, it is essential to identify and remove these attributes as well as the incomplete data from the dataset. So it is essential to preprocess data to increase data quality and get useful information. The data are preprocessed after data collection. The dataset is cleaned by discarding missing values, tuples and removing noisy values from input instances. After preprocessing, the dataset has 61,285 instances.

D. Feature Selection

Feature selection is a process of identifying essential attributes and features from the existing dataset. These selected attributes perform only relevant information for getting the expected output. Feature extraction is applied during the feature selection step to remove redundant data. Feature extraction means the selection of optimum input attributes from the selected dataset. Selected features include PM_{2.5}, PM₁₀, NO, NO₂, NO_x, NH₃, CO, SO₂, O₃, benzene, toluene, xylene, and AQI_Bucket. The number of attributes was eliminated from the selected dataset because they were not necessary for the expected results. Eliminated attributes include Station_Id and Date_Time. The target variable is AQI_Bucket. Table II is presented the summary statistics for the selected feature set.

TABLE II. SAMPLE RECORDS FOR THE SELECTED FEATURE SET

Feature	F1	F2
PM2.5	353	361.5
PM10	538	252.5
NO	101.15	119.85
NO2	142.88	156.9
NOx	158.5	181.05
NH3	51.52	62.5
CO	2.27	3.88
SO2	22.25	18.13
O3	1.28	1.4
Benzene	3.85	6.08
Toluene	88.48	104.15

Xylene	3.13	7.75
AQI_Bucket	Severe	Severe

E. Construct classification models

The dataset is classified using different classification algorithms like the decision tree algorithm [27], SVM algorithm [28], and random forest classification algorithm [29].

1. SVM: SVM is a supervised method for machine learning which is used either for classification difficulties or regression. It is utilized primarily in classification issues, though.
2. Decision Tree: One of the supervised learning methods for machine learning is the decision tree algorithm. We can utilize the decision tree algorithm for both the classification and regression analysis. The objective of this technique is to produce a model that predicts the value of a target variable for which the decision tree utilizes the representation of the tree to resolve the problem in which the leaf node matches the class label and the internal node of the tree displays characteristics.
3. Random Forest: Random Forest is another common machine learning technique, which is a part of the supervised learning process. Similar to decision trees, random forests too can be utilized in machine learning for both classification and regression analysis. It is built on the notion of ensemble learning, a technique that combines several classifications to resolve a complicated problem and enhance the model's performance. It contains several decision-making trees on different subsets of the data and takes the average to enhance the prediction precision of the data set. The random forest gets the forecast from each tree rather than depending on a decision tab and is based on the majority vote of the forecasts. and it predicts the final output.

Weka software is used to implement these algorithms. Weka is open-source with a large number of classification algorithms. The best model is selected by comparing output accuracy. The classified instance accuracy depends on the percentage split value. Data splitting aims to avoid model overfitting by dividing the dataset into train and test data. For example, if there are 700 instances in a data set, a 70% split is applied. It means 490 instances (70%) are needed to train the model, and the remaining 210 instances (30%) are applied during the testing phase.

IV. RESULTS AND DISCUSSION

Data preprocessing and feature selection phases described in Section III were helped to clean the dataset by removing missing values and unnecessary features. After data preprocessing, features of data can easily be interpreted by machine learning algorithms. Table III shows dataset details after cleaning.

TABLE III. DATASET SUMMARY

	Before	After
# of Instance	100,000	61,285
Dataset	Station_ID, DateTime, PM _{2.5} ,	PM _{2.5} , PM ₁₀ , NO, NO ₂ ,

Features	PM ₁₀ , NO, NO ₂ , NO _x , NH ₃ , SO ₂ , O ₃ , CO, Benzene, Toluene, Xylene, AQI, AQI Bucket	NO _x , NH ₃ , SO ₂ , O ₃ , CO, Benzene, Toluene, Xylene, AQI Bucket
Number of Features	16	13

As mentioned in the Methodology section, different machine learning algorithms were used to classify the dataset after preprocessing it. Table IV, Table V, and Table VI show the results related to the decision tree model, SVM model, and random forest model respectively.

TABLE IV. RESULTS OF THE DECISION TREE MODEL

Percentage split (%)	70	75	80	85	90
Accuracy (%)	63.32	63.75	64.94	64.12	63.71
Lost rate (%)	63.71	26.25	35.06	35.88	36.29
No. of Instance	18385	15321	12257	9193	6128
MAE	0.234	0.133	0.131	0.132	0.135

TABLE V. RESULTS OF SVM MODEL

Percentage split (%)	70	75	80	85	90
Accuracy (%)	61.41	61.42	61.34	60.62	60.31
Lost rate (%)	38.59	38.58	38.69	39.78	39.69
No. of Instance	18385	15321	12257	9193	6128
MAE	0.234	0.234	0.233	0.234	0.234

TABLE VI. RESULTS OF RANDOM FOREST MODEL

Percentage split (%)	70	75	80	85	90
Accuracy (%)	73.31	73.52	74.04	73.34	74.04
Lost rate (%)	26.69	26.48	25.96	26.66	25.96
No. of Instance	18385	15321	12257	9193	6128
MAE	0.136	0.135	0.134	0.135	0.134

Output results show that 80% data splitting percentage generates the highest accuracy among all algorithms. Therefore, 80% of instances are better for training the model, and the rest of 20% of data is used for testing. The above results prove that the most accurate classification is done by a random forest classification algorithm. It performs a maximum 74.0393% of accuracy value. But the implementation of the random forest algorithm takes much time when compared to the others, because the dataset is very large. The decision tree algorithm integrates some decisions and the random forest integrates various decision trees to get output. When the dataset is getting larger, the random forest algorithm generates more complex output and requires more time. So decision tree algorithm is better to use for a vast dataset, because it is faster and easier to operate than a random forest classification algorithm.

V. CONCLUSION

This study focuses on the air pollution and the application of several machine teaching techniques to categorize air contaminants with a lower error. This study helps to understand better trends, causes, sources of pollution, and important air pollutants in air pollution. In addition, many machine learning algorithms have been employed, and their abilities and limits were discovered when used for the categorization of air pollutant concentrations. The fast expansion of IoT-based technology has led to the detection, classification, and prediction of air pollution methods. They have also become trends that lead to more precise and valuable assumptions. The results of this research prove that machine learning models can be used to detect and classify air pollutants more accurately. Moreover, output results will be important for common people as well

as the government, and environmental departments, because these results will help to assess, predict air quality, and take action to prevent and minimize the adverse effects of air pollution.

As a future direction, it is proposed to analyze the data with the LSTM approach. Further, it is possible to employ artificial intelligence techniques to predict the AQI.

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